FLUX RIGIDITY FOR ANCIENT SINGLE-MODE SOLUTIONS OF THE TWO-DIMENSIONAL HYDROSTATIC EULER EQUATIONS

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Abstract. We prove a Liouville-type flux-rigidity theorem for the two-dimensional hydrostatic Euler equations on the periodic cell $\mathbb{T}^2$. For each $k \geq 1$, the class of exact solutions whose ξ-spectrum is supported on $\{0, \pm k\}$ is exactly parametrized in Fourier form: an arbitrary smooth mean profile $b(Z, \tau)$ transports an arbitrary initial fluctuation by the accumulated shear, and the modal energy $S = P^2 + Q^2$ is time-independent. The main theorem states that every such solution which is ancient, has uniformly bounded velocity component v on $\tau \leq 0$, and for which, at almost every time, no regular stream-function level component realizes the horizontal generator of $H_1(\mathbb{T}^2; \mathbb{Z}_2)$, has vanishing backward Cesàro mean flux $\bar{\Phi} = -\overline{\langle u^2 v \rangle} = 0$. For a time-independent mean profile the mechanism is secular; in general it combines an annular separation lemma with an anchored estimate on the time-averaged shear. Each hypothesis is necessary at every fixed k: explicit witnesses carry flux $5k^2\varepsilon/8$ when the v-bound is dropped, and a compensated-gap shear when horizontal winding is allowed. A cylinder analogue holds for time-independent mean profiles; the multi-mode problem remains open.

1. Introduction

The two-dimensional hydrostatic Euler equations (also called the homogeneous hydrostatic equations) arise as the formal small-aspect-ratio limit of the 2D incompressible Euler equations and as the inviscid backbone of the Prandtl hierarchy. The rigorous theory around them is by now substantial: see Grenier [4] and Brenier [2] for the derivation of the limit and a variational (optimal-transport) approach to well-posedness and finite-time breakdown for convex profiles; Masmoudi–Wong [11] for the H^s theory under convexity, and Kukavica–Temam–Vicol–Ziane [8] and Kukavica–Masmoudi–Vicol–Wong [7] for local well-posedness in analytic and structured classes; Renardy [12] and Han-Kwan–Nguyen [5] for ill-posedness outside such classes; Wong [13] and Cao–Ibrahim–Nakanishi–Titi [3] for finite-time blowup of smooth solutions; and Boutros–Markfelder–Titi [1] for the modern weak-solution theory, including nonuniqueness of generalised weak solutions. On the periodic cell $\mathbb{T}^2 \ni (\xi, Z)$ the system reads, for a stream function $W(\xi, Z, \tau)$ with *anisotropic vorticity* $w := W_{\xi\xi}$,

$$w_\tau + \{W, w\} = 0, \qquad \{A, B\} := A_\xi B_Z - A_Z B_\xi, \tag{1}$$

equivalently (see §2.2)

$$u_\tau + v u_\xi + u u_Z = -p_Z, \qquad p_\xi = 0, \qquad u = W_\xi, \quad v = -W_Z, \quad v_\xi + u_Z = 0.$$

(Our variable names are adapted to the application below: ξ is the periodic "fast" direction carrying the wave, Z the transverse direction carrying the shear.)

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Because the pressure is independent of ξ, the mean profile $b := \langle W\rangle_\xi$ has *no prognostic equation*: averaging the momentum equation in ξ only pins the pressure gradient, $p_Z = -\partial_Z\langle u^2\rangle_\xi$, and b enters the dynamics of the fluctuation only through the mean shear $V := b_Z$. We call this the *undetermined mean sector*; it is the source of both the interest and the difficulty of the problem studied here. (It is not a gauge freedom: changing b_Z changes the physical velocity $v = -W_Z$. The only true gauge freedom of the stream function is $W \mapsto W + \beta_0(\tau)$, which leaves (u, v) unchanged; see §2.3.)

The quantity of interest is the *flux*

$$\Phi := \langle W_Z W_\xi^2\rangle = -\langle u^2 v\rangle, \tag{2}$$

where $\langle\cdot\rangle$ denotes the normalized average over $\mathbb{T}^2$ (§2.1), together with its *ancient mean*

$$\bar\Phi := \lim_{T\to\infty}\frac{1}{T}\int_{-T}^{0}\Phi(\tau)\,d\tau \tag{3}$$

(a backward Cesàro mean; on the classes treated below the limit is shown to exist together with its value). The flux is the pairing of the undetermined mean shear with the modal energy weight of the wave; its status as an observable, the normalization of the velocity field, and its behavior under Galilean boosts are discussed in Remarks 2.1 and 2.2.

We work in the genre of Liouville theorems for *ancient bounded* solutions, in the spirit of Koch–Nadirashvili–Seregin–Šverák [6] for the Navier–Stokes equations and of the axisymmetric rigidity theorem of Lei–Zhang [9]: one restricts attention to solutions defined on the whole backward time axis $\tau \in (-\infty, 0]$ and satisfying uniform bounds, and asks which transport can survive. Two hypotheses are used; they are exactly those consumed by the proofs below, and each is shown — at every fixed harmonic — to be individually non-removable in Section 7. (We do not claim minimality against every conceivable weakening: for instance the uniform bound on v could in principle be relaxed to a direct control of the accumulated phase; the point is that neither hypothesis can be dropped outright.)

Hypothesis 1.1 (Ancient v-bound)**.** W is a solution of (1) on $\tau \in (-\infty, 0]$ with

$$C_v := \sup_{\tau\le 0}\|v(\cdot,\cdot,\tau)\|_{L^\infty} < \infty.$$

Hypothesis 1.2 (No horizontal winding)**.** For a.e. $\tau \le 0$ (with respect to Lebesgue measure) and every regular value c of $W(\cdot,\cdot,\tau)$ — that is, every c outside the null set of critical values furnished by Sard's theorem — no connected component of the level set $\{W(\cdot,\cdot,\tau) = c\}$ realizes the horizontal generator of $H_1(\mathbb{T}^2;\mathbb{Z}_2)$.

Hypothesis 1.2 forbids exactly one topological event: a regular level component winding around the torus in the ξ-direction. Vertical winding, and arbitrary behavior at critical values, are allowed; the conventions are discussed in Remark 2.3.

Main results. The object of this paper is the *single-mode stratum of harmonic k*: solutions whose ξ-spectrum lies in $\{0, \pm k\}$, for a fixed integer $k \ge 1$. Our first result (Lemma 3.1) is that each such class is closed under the evolution and admits an exact parametrization in Fourier form:

$$W \;=\; b(Z,\tau) + P_0(Z)\cos\bigl(k\xi + k\,\partial_Z B(Z,\tau)\bigr) + Q_0(Z)\sin\bigl(k\xi + k\,\partial_Z B(Z,\tau)\bigr),$$

with $B(Z,\tau) := \int_0^\tau b(Z,s)\,ds$: the fluctuation at time τ is the initial fluctuation composed with the accumulated shear, $W'(\xi,Z,\tau) = W_0'(\xi + \partial_Z B(Z,\tau), Z)$, uniformly in k. In particular the *modal energy* $S := P^2 + Q^2 = P_0^2 + Q_0^2$ is time-independent, while the mean profile b

remains completely free (the undetermined mean sector). On this class the flux is a pure mean-shear channel:

$$\Phi(\tau) = \frac{k^2}{4\pi} \oint b_Z(Z,\tau)\, S(Z)\, dZ.$$

Theorem 1.3 (Flux rigidity on the single-mode strata)**.** *Fix $k \geq 1$ and let W be a single-mode solution of harmonic k satisfying Hypotheses 1.1 and 1.2. Then the ancient mean flux exists and vanishes: $\bar{\Phi} = 0$.*

Corollary 1.4 (KNSS-type form)**.** *The conclusion holds, in particular, for every single-mode solution which is ancient with* both *velocity components uniformly bounded and* confined, *i.e. such that for a.e. τ every regular level component of $W(\cdot,\cdot,\tau)$ is contractible in $\mathbb{T}^2$.*

Indeed, a contractible component is null-homotopic, hence trivial in $H_1(\mathbb{T}^2;\mathbb{Z}_2)$, so confinement implies Hypothesis 1.2; and boundedness of both components implies Hypothesis 1.1. We emphasize that the corollary's hypotheses are strictly stronger than the theorem's: on the single-mode class the u-bound is automatic ($\sup_\xi |u| = k\sqrt{S}$ is finite and time-independent by Lemma 3.1), and full contractibility excludes even the standing wave $W = \sin(k\xi)$, whose regular levels are vertical circles, while Hypothesis 1.2 admits it. The witnesses of Section 7 show that Hypotheses 1.1 and 1.2 are each individually necessary, at every fixed harmonic (Remark 7.4).

The proof splits into two regimes: the separation topology that enforces Hypothesis 1.2 is common to both (Remark 5.8), but the analytic obstruction differs. (The one-line assembly of Theorem 1.3 is given at the end of Section 5.)

Time-independent mean profile ($b = b(Z)$; Theorem 4.1). Here the evolution is a shear conjugacy, level topology is frozen in time, and the mechanism is *secular*: the velocity component v contains a term of amplitude $k\sqrt{S(Z)}\,|b''(Z)|\,|\tau|$, so any Z with $S(Z)b''(Z) \neq 0$ makes $\sup|v|$ grow linearly in backward time — **flux costs unbounded** v. The velocity bound therefore forces $b'' \equiv 0$ on $\{S > 0\}$, Hypothesis 1.2 forces $b' \equiv 0$ on the interior of $\{S = 0\}$ (a horizontal circle in a spectral gap realizes the horizontal generator), and a density-plus-continuity argument propagates $b'' \equiv 0$ to all of $\mathbb{T}$, whence $b' \equiv c_0$ and finally $c_0 = 0$.

Time-dependent mean profile (Theorem 5.5). Here the conjugacy structure is lost and level topology is *not* frozen. A serious candidate configuration exists at the formal level: backward-sharpening averaged-shear fronts located at simple zeros of $\sqrt{S}$, which satisfy the v-bound by exact saturation and arrange their zero circles to be critical — mean shear dominated by $|X_0'|$ there — hence exempt from Hypothesis 1.2 (a front whose zero circles were instead regular would violate the hypothesis outright). The theorem is nevertheless true, by a different mechanism which we isolate as an *annular separation* argument (Lemmas 5.3 and 5.4): on the closed annulus between two consecutive zeros of $\sqrt{S}$ the stream function is constant on each boundary circle; if the two boundary values differ, every intermediate Sard-regular level separates the ends of the annulus, and a finite union of components none of which carries the core class cannot separate — so some component realizes the horizontal generator, violating Hypothesis 1.2. This pins the mean profile, $b(z_{j+1},\tau) = b(z_j,\tau)$ for every τ, on every arc between consecutive zeros of $\sqrt{S}$. An anchored estimate derived from the v-bound, $k\sqrt{S}\,|\partial_Z \bar{V}_T| \leq K_0/T$ with $K_0 = 2C_v + \|(P_0', Q_0')\|_\infty$, for the backward time-average $\bar{V}_T$ of the shear, then forces the averaged shear to equidistribute to zero against the weight S, and $\bar{\Phi} = 0$ follows.

The two hypotheses are each necessary, at every fixed harmonic (Section 7): dropping the v-bound admits the explicit flux-carrying witness of Proposition 7.1, and allowing horizontal winding admits a compensated-gap shear witness (Proposition 7.2). The theorem for time-independent mean profiles has a cylinder analogue (Theorem 6.1); the optimal cylinder statement for time-dependent mean profiles is left open (Remark 6.2).

Context and scope. Steady cells ($b_Z \equiv 0$) and traveling waves are the classical zero-flux members of the ancient class. For *steady* hydrostatic flows rigidity results are available: Leung–Wong–Xie [10] prove, as one ingredient of their characterization of steady nozzle flows, that steady solutions in an infinite strip strictly away from stagnation are shear flows (see also the references there). The present result concerns the genuinely time-dependent ancient regime, where the freedom of the undetermined mean sector is the central difficulty; with the classification stated in Fourier form, the theorem covers each single-mode stratum in full. The multi-mode problem remains open — to the best of the author's knowledge, no Liouville-type result is available for the time-dependent ancient regime beyond the single-mode strata — and appears to be of a different nature; we state it precisely in Section 8. We do not claim any consequence at the level of the primitive (axisymmetric Navier–Stokes) equations.

A remark on verification: the classification identities of Lemma 3.1, the flux formula, and the counterexample's flux value are verified in exact (symbolic) arithmetic, and the counterexample's residual and secular growth in float64 grid arithmetic, by short standalone scripts that accompany this paper as ancillary files (Appendix A); none is load-bearing for the proofs, which are complete in the text. A plausible-but-false variant of one step, worth flagging for arguments of this genre, is discussed in Remark 4.5.

2. Setting

Throughout, $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$ and $\mathbb{T}^2 \ni (\xi, Z)$; all functions are smooth unless stated otherwise (for the time-independent mean profile, C^2 regularity suffices; see Remark 4.4).

2.1. **Conventions.** Averages are normalized:

$$\langle f\rangle := \frac{1}{4\pi^2}\int_0^{2\pi}\int_0^{2\pi} f(\xi,Z)\,d\xi\,dZ, \qquad \langle f\rangle_\xi := \frac{1}{2\pi}\int_0^{2\pi} f(\xi,Z)\,d\xi,$$

and $\oint$ denotes $\int_0^{2\pi}\cdot\,dZ$. The system is (1) with $u = W_\xi$, $v = -W_Z$, $w = W_{\xi\xi} = u_\xi$. The flux is (2); note that $\langle W_Z W_\xi^2\rangle = -\langle u^2 v\rangle$ holds pointwise, since $v = -W_Z$ (both forms will be used).

2.2. **Equivalence with the momentum form; pressure recovery.** Given a solution of the momentum form, differentiating $u_\tau + vu_\xi + uu_Z = -p_Z$ in ξ eliminates the pressure and, using $v_\xi + u_Z = 0$, yields (1) for $w = u_\xi$. Conversely, given a solution W of (1), average the momentum equation in ξ: since $\langle u\rangle_\xi \equiv 0$ and $\langle vu_\xi + uu_Z\rangle_\xi = \partial_Z\langle u^2\rangle_\xi$ (integrate the first term by parts in ξ and use $v_\xi = -u_Z$), the pressure is recovered, up to a function of time, by

$$p_Z = -\partial_Z\langle u^2\rangle_\xi,$$

and the fluctuating part of the momentum equation is the ∂_ξ-antiderivative of (1). This justifies "equivalently" above.

2.3. **The undetermined mean sector; frame normalization.** Decomposing $W = b(Z,\tau) + W'$ with $\langle W'\rangle_\xi = 0$: there is no evolution equation for b; the mean profile is a free trajectory of the undetermined mean sector, constrained only through its interaction with the fluctuation, which happens exclusively through the mean shear $V = b_Z$. The true gauge freedom is $W \mapsto W + \beta_0(\tau)$, which leaves (u,v) — and hence Φ and $\bar\Phi$ — unchanged.

Remark 2.1 (Circulations and Galilean frame). The velocity field of (1) is $(\dot{\xi}, \dot{Z}) = (v, u)$. On $\mathbb{T}^2$ a general divergence-free field carries two harmonic (constant) components; the representation by a *periodic* stream function fixes them to zero: $\langle u \rangle = \langle W_\xi \rangle = 0$ and $\langle v \rangle = -\langle W_Z \rangle = 0$. The setting thus selects the canonical rest frame of zero mean velocity. A Galilean boost along ξ acts by $\widetilde{W}(\xi, Z, \tau) = W(\xi - c\tau, Z, \tau) - cZ$, so that $\widetilde{v} = v(\xi - c\tau, \cdot, \cdot) + c$ and $\widetilde{u} = u(\xi - c\tau, \cdot, \cdot)$ and the equation is preserved; the boosted stream function is no longer Z-periodic, and the flux shifts by

$$\Phi \longmapsto \Phi - c\,\langle u^2 \rangle,$$

with $\langle u^2 \rangle = \frac{k^2}{4\pi} \oint S\,dZ$ conserved on the classes below; the value of Φ is therefore frame-dependent, and Theorem 1.3 is a statement in the canonical rest frame.

Remark 2.2 (The flux as an observable). The flux (2) is not the transport of a conserved density: the ξ-averaged budget of the (hydrostatic) kinetic energy reads, for every solution,

$$\partial_\tau \langle \tfrac{1}{2} u^2 \rangle_\xi + \partial_Z \langle \tfrac{1}{2} u^3 \rangle_\xi = 0$$

(multiply the momentum equation by u, average in ξ, and use $p_\xi = 0$, $\langle u \rangle_\xi = 0$, $v_\xi = -u_Z$), and on the single-mode class both terms vanish separately ($\langle u^2 \rangle_\xi = \frac{k^2}{2} S$ is time-independent and $\langle u^3 \rangle_\xi \equiv 0$ by parity). Rather, on the class the flux reduces to the pairing $\Phi = \langle V \, \langle u^2 \rangle_\xi \rangle$ of the *undetermined* mean shear with the conserved modal weight (Lemma 3.4): it is the natural scalar diagnostic of the mean sector against the wave, and since b obeys no equation, a constraint on $\bar{\Phi}$ is genuinely a constraint on which mean trajectories the class admits, not a consequence of a closed balance law. The theorem shows that, for ancient solutions satisfying Hypotheses 1.1 and 1.2, no persistent correlation between the mean shear and the modal weight can survive in the Cesàro mean.

Remark 2.3 (Conventions in Hypothesis 1.2). (i) The quantifier is over *components*: a level set may have several components, and the hypothesis constrains each one separately (replacing "component" by "union of components" would give a different, and for our purposes wrong, condition). (ii) The hypothesis is stated per regular *value*; this single reading is used at both consumption sites (Step 3 of Theorem 4.1, Lemma 5.4). Under the alternative per-*component* reading — forbidding horizontal winding for every component on which $\nabla W \neq 0$, regardless of the value — the theorems also hold, with slightly lower regularity requirements (Remark 4.4). (iii) Only the $\mathbb{Z}_2$-class of the horizontal generator is used, which is why the hypothesis is phrased in $H_1(\mathbb{T}^2; \mathbb{Z}_2)$.

3. The single-mode classes

Fix an integer $k \geq 1$.

Lemma 3.1 (Classification; Fourier form)**.** *Let W be smooth on $\mathbb{T}^2 \times (-\infty, 0]$ with ξ-spectrum in $\{0, \pm k\}$ for every τ, and write*

$$W = b(Z, \tau) + P(Z, \tau)\cos(k\xi) + Q(Z, \tau)\sin(k\xi).$$

Then W solves (1) if and only if

$$\partial_\tau (P, Q) = k\, b_Z\, (Q, -P), \tag{4}$$

with the mean profile b unconstrained (the undetermined mean sector of §2.3). Equivalently, with $B(Z, \tau) := \int_0^\tau b(Z, s)\, ds$ and $(P_0, Q_0) := (P, Q)(\cdot, 0)$,

$$W = b(Z, \tau) + P_0(Z)\cos\big(k\xi + k\,\partial_Z B(Z, \tau)\big) + Q_0(Z)\sin\big(k\xi + k\,\partial_Z B(Z, \tau)\big) : \tag{5}$$

the fluctuation at time τ is the initial fluctuation composed with the accumulated shear, $W'(\xi, Z, \tau) = W_0'\big(\xi + \partial_Z B(Z,\tau), Z\big)$, uniformly in k. In particular the modal energy

$$S(Z) := P^2 + Q^2 = P_0^2 + Q_0^2$$

is time-independent, and (5) *is* exactly *the single-mode class of harmonic k: every such solution is realized by exactly one pair of smooth 2π-periodic Fourier coefficients (P_0, Q_0) and one smooth mean trajectory $b(\cdot,\cdot)$, namely its own.*

Proof. With $W' = P\cos(k\xi) + Q\sin(k\xi)$ one has $w = W_{\xi\xi} = -k^2 W'$, so $\{W', w\} = -k^2\{W', W'\} = 0$ identically: the fluctuation self-interaction of a single harmonic is null (no $m = \pm 2k$ modes are generated, and the class is closed under the evolution). Hence

$$\{W, w\} = \{b, w\} = -b_Z\,\partial_\xi w = k^3\, b_Z\,\big(Q\cos(k\xi) - P\sin(k\xi)\big),$$

while $w_\tau = -k^2\big(P_\tau\cos(k\xi) + Q_\tau\sin(k\xi)\big)$; matching the $\cos(k\xi)$ and $\sin(k\xi)$ coefficients of $w_\tau + \{W, w\} = 0$ gives exactly (4), and the $m = 0$ sector of the vorticity equation is empty, leaving b free. Writing $\zeta := P + iQ$, (4) reads $\zeta_\tau = -ik\, b_Z\,\zeta$, whence $\zeta(Z,\tau) = e^{-ik\,\partial_Z B(Z,\tau)}\,\zeta_0(Z)$ (note $\partial_Z B = \int_0^\tau b_Z$), which is (5); in particular $S = |\zeta|^2$ is conserved. Conversely, (5) defines an exact solution for every smooth datum by the same computation. Uniqueness of the realization is immediate: $b = \langle W\rangle_\xi$ and (P_0, Q_0) are the Fourier coefficients of $W(\cdot,\cdot,0)$. The identities of this proof are additionally machine-verified in exact symbolic arithmetic (Appendix A). □

Lemma 3.2 (Time-independent mean profile; shear conjugacy)**.** *If $b = b(Z)$ is time-independent,* (5) *becomes*

$$W = b(Z) + P_0(Z)\cos\big(k\xi + k\,b'(Z)\,\tau\big) + Q_0(Z)\sin\big(k\xi + k\,b'(Z)\,\tau\big) \;=\; W_0 \circ H_\tau, \tag{6}$$

where $H_\tau(\xi, Z) = (\xi + b'(Z)\tau,\, Z)$: the evolution is a shear conjugacy by a diffeomorphism of $\mathbb{T}^2$ isotopic to the identity; in particular the component topology of every level set is frozen in time.

Proof. Immediate from (5) with $\partial_Z B(Z,\tau) = b'(Z)\tau$. □

Remark 3.3 (Polar form; two obstructions to a global amplitude–phase pair). Where a smooth 2π-periodic pair (A, φ_0) with $W_0' = A(Z)\sin\big(k\xi + k\varphi_0(Z)\big)$ exists, (5) takes the amplitude–phase form

$$W = b(Z,\tau) + A(Z)\,\sin\big(k\xi + k\varphi_0(Z) + k\,\partial_Z B(Z,\tau)\big), \qquad A^2 = S, \tag{7}$$

which we use below to write down explicit witnesses (Proposition 7.1, Remark 5.9, Proposition 7.2). But not every single-mode field admits such a pair, for two distinct reasons, so (7) parametrizes a *proper* subclass of the stratum. (i) *Phase winding*: for $W = \cos(\xi - Z)$ — a smooth steady $k = 1$ solution with spectrum $\{\pm 1\}$ — any polar phase equals $\pi/2 - Z$ modulo 2π and sign, which has winding -1 in Z and admits no 2π-periodic real-valued choice; and since $\varphi_0 + \partial_Z B$ is a sum of 2π-periodic functions, the form (7) can never produce a winding phase. (ii) *Sign lifting through zeros*: for $(P_0, Q_0) = \cos(Z/2)\,\big(\cos(Z/2), \sin(Z/2)\big)$ (a $k = 1$ datum), smooth and 2π-periodic with $S = (1 + \cos Z)/2$, any smooth signed amplitude equals $\pm\cos(Z/2)$ near the simple zero at $Z = \pi$, and C^1-matching there forces one global sign, which is incompatible with periodicity ($A(0) = -A(2\pi)$); only an *antiperiodic* pair represents this field. For these reasons the classification above, the flux formula, and every estimate in Sections 4–6 are stated and proved in the representation-free variables (P_0, Q_0), S, $\partial_Z B$ only.

Lemma 3.4 (Flux formula)**.** *On the class* (5),

$$\langle W_Z u^2\rangle_\xi = \tfrac{k^2}{2}\, b_Z(Z,\tau)\, S(Z) \qquad \Longrightarrow \qquad \Phi(\tau) = \frac{k^2}{4\pi}\oint b_Z(Z,\tau)\, S(Z)\, dZ. \tag{8}$$

In the decomposition of the flux into mean-shear and fluctuation channels, $\Phi = \langle V\,\langle u^2\rangle_\xi\rangle + \langle W'_Z u^2\rangle$ *with* W' *the* ξ*-fluctuation of* W*, the fluctuation channel vanishes identically on the class by* ξ*-parity: the single-mode flux is a pure mean-shear channel. For a time-independent mean profile* (6)*,* Φ *is time-independent and* $\bar\Phi = \frac{k^2}{4\pi}\oint b' S\, dZ$*.*

Proof. $u = W'_\xi = k\big(-P\sin(k\xi) + Q\cos(k\xi)\big)$, so $\langle u^2\rangle_\xi = \frac{k^2}{2}(P^2+Q^2) = \frac{k^2}{2}S$ and $\langle W_Z u^2\rangle_\xi = b_Z\langle u^2\rangle_\xi + \langle W'_Z u^2\rangle_\xi$. In the second term, $W'_Z = P_Z\cos(k\xi) + Q_Z\sin(k\xi)$ multiplies $u^2 = k^2\big(P^2\sin^2 - 2PQ\sin\cos + Q^2\cos^2\big)(k\xi)$, producing only odd trigonometric moments ($\langle\cos^3\rangle = \langle\sin^3\rangle = \langle\sin\cos^2\rangle = \langle\sin^2\cos\rangle = 0$ over a period), so it vanishes; the first term is $\frac{k^2}{2}b_Z S$, which gives (8) after averaging over Z (S is time-independent by Lemma 3.1). ☐

4. The time-independent mean profile

Theorem 4.1 (Single-mode strata, time-independent mean profile; "flux costs unbounded v")**.** *Let* W *be a single-mode solution* (6) *with time-independent mean profile, satisfying Hypotheses 1.1 and 1.2. Then* $b' \equiv 0$ *on* $\mathbb{T}$*: the mean profile is constant. In particular* $b'S \equiv 0$ *and* $\bar\Phi = 0$*.*

Proof. Step 1 (the secular term). Write $X_0 := (P_0, Q_0)$, $JX_0 := (Q_0, -P_0)$, and $\zeta_0 := P_0 + iQ_0$. From (6),

$$\begin{aligned} v = -W_Z &= -\,b'(Z) - \big(P_Z\cos(k\xi) + Q_Z\sin(k\xi)\big),\\ \sup_\xi\big|P_Z\cos(k\xi) + Q_Z\sin(k\xi)\big| &= \big|X_0'(Z) + k\,\tau\, b''(Z)\, JX_0(Z)\big|, \end{aligned} \tag{9}$$

where the identity on the second line holds because $\zeta := P + iQ = e^{-ikb'(Z)\tau}\zeta_0$ (Lemma 3.1) gives $\zeta_Z = e^{-ikb'\tau}\big(\zeta_0' - ik\tau b''\zeta_0\big)$, the modulus is rotation-invariant, and $\zeta_0' - ik\tau b''\zeta_0$ is the complex form of $X_0' + k\tau b'' JX_0$. Every contribution to v is bounded uniformly in τ except the secular one: since $|JX_0| = \sqrt{S}$,

$$\sup_{\mathbb{T}^2}|v(\cdot,\cdot,\tau)| \;\geq\; k\,|\tau|\,|b''(Z)|\sqrt{S(Z)} - |X_0'(Z)| - |b'(Z)| \qquad \text{for every } Z.$$

If $S(Z)b''(Z) \neq 0$ for some Z, the right side tends to ∞ as $\tau \to -\infty$, contradicting Hypothesis 1.1. Hence

$$S(Z)\, b''(Z) \equiv 0 \quad \text{on } \mathbb{T}.$$

Step 2 (structure of b' *on the wave support).* Let $\Omega := \{Z : S(Z) > 0\}$, an open subset of the circle. By Step 1, $b'' \equiv 0$ on Ω, so b' is constant on each connected component of Ω.

Step 3 (spectral gaps). Let $I \subseteq \operatorname{int}\{S = 0\}$ be a nonempty open interval (a *gap*; int denotes the interior of a set). On $\mathbb{T}_\xi \times I$ the solution is $W = b(Z)$: no ξ-dependence. Suppose $b'(Z_0) \neq 0$ for some $Z_0 \in I$. Pick a closed subinterval $J \subset I$ with $b' \neq 0$ on J; then $b(J)$ is a nondegenerate interval, and by Sard's theorem a.e. $c^* \in b(J)$ is a regular value of $W(\cdot,\cdot,\tau)$. For any such $c^* = b(Z^*)$, $Z^* \in J$, the level component through $(\cdot, Z^*)$ is exactly the horizontal circle $\{Z = Z^*\}$: regular (since $|\nabla W| = |b'(Z^*)| \neq 0$ there) and realizing the horizontal generator of $H_1(\mathbb{T}^2;\mathbb{Z}_2)$. The gap region is invariant under the shear conjugacy of Lemma 3.2, so this component persists for all τ, violating Hypothesis 1.2. Hence $b' \equiv 0$ on $\operatorname{int}\{S = 0\}$ (and so also $b'' \equiv 0$ there).

Step 4 (density). By Steps 2 and 3, $b''=0$ on the open set $\Omega\cup\operatorname{int}\{S=0\}$, which is *dense* in $\mathbb{T}$: its complement is $\{S=0\}\setminus\operatorname{int}\{S=0\}$, every point of which is a limit of points of Ω. Since b'' is continuous, $b''\equiv 0$ on $\mathbb{T}$, so $b'\equiv c_0$ is a global constant. If a gap exists, $c_0=0$ directly by Step 3; if $\{S=0\}$ has empty interior, periodicity gives $\oint b'\,dZ=0$, hence again $c_0=0$, i.e. $b'\equiv 0$. In particular $b'S\equiv 0$ and, by Lemma 3.4, $\bar\Phi=\frac{k^2}{4\pi}\oint b'S\,dZ=0$. □

Remark 4.2 (Mechanism). Nonzero flux needs $\oint b'S\neq 0$; under Hypothesis 1.2 the shear b' must live on $\{S>0\}$; but any non-constancy of b' there is paid for by the secular term of size $k\sqrt{S}\,|b''|\,|\tau|$ in v — the configuration forfeits its own boundedness. We call this mechanism *flux costs unbounded* v.

Remark 4.3 (Where each hypothesis bites). In the time-independent case, the v-bound is used *only* through the secular term (Step 1), Hypothesis 1.2 *only* in the gaps (Step 3) — if $\{S=0\}$ has empty interior, Hypothesis 1.2 is not needed at all — and periodicity *only* for $\oint b'=0$ in the gap-free branch of Step 4. In the time-dependent case, the v-bound enters only through the anchored cap (Lemma 5.2) and Hypothesis 1.2 only through the annular separation (Lemma 5.4). On the cylinder, by contrast, boundedness of u is what supplies the boundedness of S ($\sup_\xi|u|=k\sqrt{S}$) consumed by the properness argument of Theorem 6.1; on the torus S is bounded automatically. Each hypothesis is individually necessary at every fixed harmonic (Section 7).

Remark 4.4 (Regularity). For Theorem 4.1, $W(\cdot,\cdot,0)\in C^2$ suffices: $b\in C^2$, $P_0,Q_0\in C^2$ (so that two-dimensional Sard applies to W and the per-value reading of Step 3 is licensed; under the per-component reading of Remark 2.3, where the gap circle $\{Z=Z^*\}$ is exhibited directly as a component with $\nabla W\neq 0$, $P_0,Q_0\in C^1$ suffices), with b'' continuous for Steps 2 and 4. Elsewhere (the time-dependent mean profile and the Sard selections) smoothness is the standing assumption and we do not claim an optimal threshold.

Remark 4.5 (A plausible-but-false adjacency variant). In Step 4 it is tempting to argue instead that each component of Ω is adjacent to a gap and to chain the constants b' through; this is false in general: a component of Ω may abut only nowhere-dense zeros of $\sqrt{S}$, with no gap on either side (e.g. Cantor-type zero sets), and chains through isolated zeros need not terminate at a gap. The density argument avoids any case analysis on the structure of the zero set. We flag the variant because plausible adjacency claims about the components of an open subset of the circle seem a recurring hazard for arguments of this genre.

5. The full theorem: time-dependent mean profile

For a time-dependent mean profile $b(Z,\tau)$ the class is (5). Two structural facts change at once: the evolution is no longer a conjugacy (level topology is *not* frozen), and the secular mechanism of Theorem 4.1 is no longer available (the phase now carries $\partial_Z B$, whose growth the mean trajectory can shape at will). A serious candidate configuration exists at the formal level: backward-sharpening averaged-shear fronts concentrated at simple zeros of $\sqrt{S}$, which satisfy the v-bound by exact saturation of the anchored estimate below and arrange the zero circles $\{Z=z_j\}$ to be critical ($|b_Z|\le|X_0'|$ there), hence exempt from Hypothesis 1.2; if instead $|b_Z|>|X_0'|$ on a zero circle, that circle is itself a regular level component realizing the horizontal generator and the hypothesis already fails (proof of Lemma 5.4). The theorem excludes this candidate not at the zero circles but *next to* them: the separating levels at values strictly between the boundary plateaus are regular, and they wind.

Throughout this section W is a member of (5) satisfying Hypotheses 1.1 and 1.2; C_v is the bound of Hypothesis 1.1; $X_0 := (P_0, Q_0)$, $S = |X_0|^2$, and

$$\bar{V}_T(Z) := \frac{1}{T}\int_{-T}^{0} b_Z(Z,\tau)\,d\tau$$

is the backward time-averaged shear at horizon $T > 0$. We assume $\sqrt{S}$ has at least one zero; otherwise $\Omega = \mathbb{T}$ and the (easier) gap-free argument of Remark 5.6 applies verbatim. (If $S \equiv 0$ the solution is a pure mean trajectory, $\Phi \equiv 0$, and there is nothing to prove.)

Lemma 5.1 (ξ-average of v; boundedness of the shear)**.** *$\langle v\rangle_\xi = -b_Z(Z,\tau)$; hence $|b_Z| \le C_v$ pointwise, and $|\bar{V}_T| \le C_v$ for every T.*

Proof. $v = -W_Z = -b_Z - \partial_Z\big(P\cos(k\xi) + Q\sin(k\xi)\big)$, and the ξ-average of the second term vanishes. □

Lemma 5.2 (Anchored cap)**.** *Let $K_0 := 2C_v + \|X_0'\|_\infty$, where $\|X_0'\|_\infty = \sup_Z\big|(P_0'(Z), Q_0'(Z))\big|$. For all Z and $T > 0$,*

$$k\sqrt{S(Z)}\,\big|\partial_Z \bar{V}_T(Z)\big| \;\le\; \frac{K_0}{T}.$$

In particular $\sqrt{S}\,\partial_Z\bar{V}_T \to 0$ uniformly as $T \to \infty$.

Proof. Write $\zeta = P + iQ$ and $\zeta_0 = P_0 + iQ_0$, so $\zeta = e^{-ik\,\partial_Z B}\zeta_0$ by Lemma 3.1 and

$$\zeta_Z = e^{-ik\,\partial_Z B}\big(\zeta_0' - ik\,\partial_Z^2 B\,\zeta_0\big), \qquad \text{whence} \qquad k\,|\zeta_0|\,\big|\partial_Z^2 B\big| \;\le\; |\zeta_Z| + |\zeta_0'|$$

by the triangle inequality ($|\zeta_Z| = |\zeta_0' - ik\,\partial_Z^2 B\,\zeta_0| \ge k|\partial_Z^2 B|\,|\zeta_0| - |\zeta_0'|$). Since $v = -b_Z - (P_Z\cos(k\xi) + Q_Z\sin(k\xi))$ and $\sup_\xi|P_Z\cos(k\xi) + Q_Z\sin(k\xi)| = |\zeta_Z|$, Hypothesis 1.1 and Lemma 5.1 give $|\zeta_Z| \le \sup_\xi|v| + |b_Z| \le 2C_v$, so

$$k\sqrt{S}\,\big|\partial_Z^2 B(Z,\tau)\big| \;\le\; 2C_v + \|X_0'\|_\infty = K_0 \qquad \text{for all } (Z,\tau).$$

Finally $B(Z,0) = 0$, so $\partial_Z\bar{V}_T = -\,\partial_Z^2 B(Z,-T)/T$, and the pointwise bound at the single anchor time $-T$ gives the claim. □

Lemma 5.3 (Separation in an annulus; topological form)**.** *Let $\mathcal{A} = \mathbb{T}_\xi \times [z_0, z_1]$ be a closed annulus and let $L \subset \mathbb{T}_\xi \times (z_0, z_1)$ be a compact embedded 1-manifold (finitely many disjoint circles) which* separates *the two boundary circles of $\mathcal{A}$: every continuous path in $\mathcal{A}$ from $\mathbb{T}_\xi \times \{z_0\}$ to $\mathbb{T}_\xi \times \{z_1\}$ meets L. Then some component of L represents the core class $[\mathbb{T}_\xi \times \{\mathrm{pt}\}] \in H_1(\mathcal{A};\mathbb{Z}_2)$; under the inclusion $\mathcal{A} \hookrightarrow \mathbb{T}^2$ this class maps to the horizontal generator of $H_1(\mathbb{T}^2;\mathbb{Z}_2)$.*

Proof. $H_1(\mathcal{A};\mathbb{Z}_2) \cong \mathbb{Z}_2$ is generated by the core circle, and $H_1(\mathcal{A},\partial\mathcal{A};\mathbb{Z}_2) \cong \mathbb{Z}_2$ is generated by the class $[\gamma]$ of any arc γ joining the two boundary circles; the mod-2 intersection pairing

$$H_1(\mathcal{A};\mathbb{Z}_2) \times H_1(\mathcal{A},\partial\mathcal{A};\mathbb{Z}_2) \longrightarrow \mathbb{Z}_2$$

is perfect (Poincaré–Lefschetz duality), with $[\mathrm{core}]\cdot[\gamma] = 1$. Let U be the connected component of $\mathcal{A} \setminus L$ containing the lower boundary circle; by separation, the upper boundary circle does not lie in U. Since $\mathcal{A}$ is orientable and each component L_i of L is an embedded two-sided circle, the two local sides of L_i lie in well-defined (possibly equal) components of $\mathcal{A} \setminus L$; call L_i *U-bounding* if exactly one of its sides lies in U. Choose γ a smooth arc from the lower to the upper boundary, transverse to L. Along γ, membership in U changes exactly at the crossings of U-bounding components (a crossing of a component with both sides in U, or with neither side in U, does not change it), and it changes an odd number of times, since γ starts in U and ends outside. Hence the total number of crossings of U-bounding components is odd, and some U-bounding component L_i has $\#(\gamma \cap L_i)$ odd, i.e. $[L_i] \cdot [\gamma] = 1$; perfectness of the pairing forces $[L_i] = [\text{core}]$ in $H_1(\mathcal{A}; \mathbb{Z}_2)$. The inclusion $\mathcal{A} \hookrightarrow \mathbb{T}^2$ sends the core circle to a horizontal circle, whose class is the horizontal generator of $H_1(\mathbb{T}^2; \mathbb{Z}_2)$. □

Lemma 5.4 (Annular separation; arc matching)**.** *Let $z_j < z_{j+1}$ be two zeros of $\sqrt{S}$ bounding a connected component (z_j, z_{j+1}) of $\Omega = \{S > 0\}$, and let $\mathcal{A} = \mathbb{T}_\xi \times [z_j, z_{j+1}]$ be the closed annulus between them. For every τ in the full-measure set of Hypothesis 1.2, the field $W(\cdot,\cdot,\tau)$ is constant on each boundary circle of $\mathcal{A}$ (equal to $\beta_j := b(z_j,\tau)$, resp. $\beta_{j+1} := b(z_{j+1},\tau)$), and $\beta_j = \beta_{j+1}$. Consequently, by continuity of $\tau \mapsto b(z_j,\tau)$,* for every $\tau \le 0$,

$$b(z_{j+1},\tau) = b(z_j,\tau) \qquad \textit{i.e.} \qquad \int_{z_j}^{z_{j+1}} b_Z(Z,\tau)\,dZ = 0. \tag{10}$$

Proof. On $\{Z = z_j\}$ we have $S = 0$, hence $P_0 = Q_0 = 0$ and $W' = 0$, so $W = b(z_j,\tau) = \beta_j$, a constant; likewise on the other boundary. Suppose $\beta_j < \beta_{j+1}$ (the other case is symmetric). By Sard's theorem a.e. $\lambda \in (\beta_j, \beta_{j+1})$ is a regular value of $W(\cdot,\cdot,\tau)$; fix such a λ. Then $W - \lambda$ takes opposite signs on the two boundary circles of $\mathcal{A}$, so the regular level $L := \{W = \lambda\} \cap \mathcal{A}$ is a nonempty compact 1-manifold (finitely many disjoint embedded circles, contained in the open annulus) which separates the two ends: every continuous path from one boundary to the other carries W from a value $< \lambda$ to a value $> \lambda$ and therefore meets L by the intermediate value theorem. By Lemma 5.3, some component of L realizes the horizontal generator of $H_1(\mathbb{T}^2; \mathbb{Z}_2)$ — and λ is a regular value, so this violates Hypothesis 1.2. Hence $\beta_j = \beta_{j+1}$ for a.e. τ, and since $\tau \mapsto b(z_{j+1},\tau) - b(z_j,\tau)$ is continuous and vanishes on a dense set, it vanishes for every τ.

Note that the critical-value exemption of Hypothesis 1.2 can protect only the zero circles $\{Z = z_j\}$ themselves: there $W_\xi \equiv 0$ and $W_Z = b_Z(z_j,\tau) + P_Z \cos(k\xi) + Q_Z \sin(k\xi)$ with $\sup_\xi |P_Z \cos(k\xi) + Q_Z \sin(k\xi)| = |X_0'(z_j)|$, so the circle carries a critical point precisely when $|b_Z(z_j,\tau)| \le |X_0'(z_j)|$ (and is then exempt, β_j being a critical value); if instead $|b_Z(z_j,\tau)| > |X_0'(z_j)|$, then W_Z has a definite sign on the circle, hence on a tubular neighborhood of it, so by the implicit function theorem every level $\{W = \lambda\}$ with λ near β_j meets the tube in a graph $\{Z = \zeta_\lambda(\xi)\}$ — a regular closed curve winding once in ξ; choosing such a λ that is moreover a Sard-regular value of W gives a regular level component realizing the horizontal generator, and Hypothesis 1.2 fails under either reading of Remark 2.3. In either case the separating levels at values strictly between β_j and β_{j+1} are regular by the Sard selection, and it is exactly these that wind. □

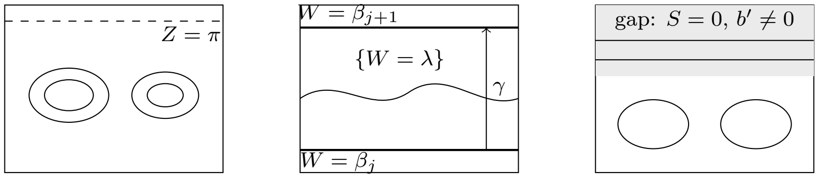
Theorem 5.5 (Full single-mode strata)**.** *Let W be any member of the class (5) satisfying Hypotheses 1.1 and 1.2. Then*

$$\lim_{T\to\infty} \oint S(Z)\,\bar{V}_T(Z)\,dZ = 0;$$

in particular the backward Cesàro mean $\bar{\Phi}$ exists and equals 0*.*

Proof. By Lemma 3.4, $\frac{1}{T}\int_{-T}^{0} \Phi\,d\tau = \frac{k^2}{4\pi}\oint S\,\bar{V}_T\,dZ$, so the displayed limit is exactly the statement that $\bar{\Phi}$ exists and vanishes.

Let $z_1 < \cdots < z_N$ be the zeros of $\sqrt{S}$ if finitely many, with the lifted endpoint convention $z_{N+1} := z_1 + 2\pi$; for $N = 1$ the single component $I_1 = (z_1, z_1 + 2\pi)$ of $\Omega = \{S > 0\}$ has both ends at the same circle, and (10) degenerates to the periodicity identity $\oint b_Z\,dZ = 0$ (the separation statement of Lemma 5.4 is vacuous there). If the zero set is infinite the argument



(a) fold ovals (b) annular separation (c) gap witness

FIGURE 1. Level-set portraits. (a) Regular levels of the counterexample of Proposition 7.1: contractible fold ovals; the seam $\{Z=\pi\}$ is entirely critical. (b) The annulus between consecutive zeros of $\sqrt{S}$: when the boundary plateaus differ, a Sard-regular intermediate level separates the ends and some component winds horizontally (Lemmas 5.3 and 5.4); γ is the transverse arc of the mod-2 pairing. (c) The compensated-gap witness of Proposition 7.2: horizontal regular levels inside the gap realize the horizontal generator.

below runs on the (at most countably many) connected components I_j of Ω, with $S\to 0$ at both ends of each component, which is all that is used. By Lemma 5.4, (10) holds for every component and *every* τ; averaging over $\tau\in[-T,0]$ gives, for every component,

$$\int_{I_j} \bar V_T(Z)\,dZ = 0. \tag{11}$$

Fix a component I_j and $\eta\in(0,|I_j|/2)$. Let collar_η denote the two subintervals of I_j within distance η of its ends, let $K_\eta := \{Z\in I_j : \mathrm{dist}(Z,\partial I_j)\ge\eta\}$ be the complementary compact subarc, and let $a_\eta := \min_{K_\eta}\sqrt{S} > 0$. By Lemma 5.2 (and $k\ge 1$), on K_η,

$$|\partial_Z \bar V_T| \le \frac{K_0}{T\,a_\eta} \qquad\Longrightarrow\qquad \mathrm{osc}_{K_\eta}\bar V_T \le \frac{K_0\,|I_j|}{T\,a_\eta} \xrightarrow[T\to\infty]{} 0,$$

where $\mathrm{osc}_K f := \sup_K f - \inf_K f$. Writing m_T for the mean of $\bar V_T$ over K_η, the zero-integral (11) and the uniform bound $|\bar V_T|\le C_v$ (Lemma 5.1) give

$$|m_T|\,|K_\eta| = \Big|\int_{K_\eta}\bar V_T\Big| = \Big|\int_{\mathrm{collar}_\eta}\bar V_T\Big| \le 2\eta C_v,$$

so $\sup_{K_\eta}|\bar V_T| \le \frac{2\eta C_v}{|I_j|-2\eta} + \mathrm{osc}_{K_\eta}\bar V_T$, and with $\|S\|_\infty := \sup_{\mathbb{T}} S$,

$$\Big|\int_{I_j} S\,\bar V_T\Big| \le \|S\|_\infty\,|I_j|\Big(\frac{2\eta C_v}{|I_j|-2\eta} + \mathrm{osc}_{K_\eta}\bar V_T\Big) + \|S\|_\infty\cdot 2\eta\,C_v.$$

Hence for every admissible η,

$$\limsup_{T\to\infty}\Big|\int_{I_j} S\,\bar V_T\Big| \le \|S\|_\infty\Big(\frac{2\eta C_v\,|I_j|}{|I_j|-2\eta} + 2\eta C_v\Big),$$

and letting $\eta\to 0$ shows $\int_{I_j} S\,\bar V_T\to 0$ for every component; outside Ω the integrand vanishes identically ($S=0$). Finally,

$$\Big|\int_{I_j} S\,\bar V_T\Big| \le C_v\,\|S\|_\infty\,|I_j|, \qquad \sum_j |I_j| \le 2\pi,$$

so dominated convergence for the series over components yields $\oint S\,\bar V_T\,dZ\to 0$. $\square$

Proof of Theorem 1.3*.* By Lemma 3.1, every single-mode solution of harmonic k is a member of (5), so Theorem 5.5 applies verbatim; time-independent mean profiles are the special case $b_\tau \equiv 0$, in which Theorem 4.1 sharpens the conclusion to $b' \equiv 0$. □

Remark 5.6 (Gap-free case)*.* If $\sqrt{S}$ has no zeros, (11) is replaced by the periodicity identity $\oint \bar V_T\, dZ = 0$, and the argument runs with $K_\eta = \mathbb{T}$, $a = \min \sqrt{S} > 0$: the anchored cap gives $\operatorname{osc}_{\mathbb{T}} \bar V_T \to 0$, the zero integral forces the near-constant to 0, and $\oint S\, \bar V_T \to 0$ directly.

Remark 5.7 (A saturating candidate at simple zeros)*.* The backward-sharpening front candidate concentrated shear growth at a simple zero z_j of $\sqrt{S}$, exactly saturating Lemma 5.2 (which alone therefore cannot exclude it), and its zero circles are arranged to be critical ($|b_Z| \le |X_0'|$ there), hence exempt from Hypothesis 1.2. It is excluded by Lemma 5.4: the exemption protects the zero circles only, while the separating levels at values strictly between the boundary plateaus are regular and wind. Hypothesis 1.2 acts here through *global separation*, not through local winding at the front.

Remark 5.8 (One separation device, two analytic drivers)*.* The horizontal-generator separation argument appears three times in this paper: in the spectral gaps (Step 3 of Theorem 4.1 is the degenerate case where a boundary plateau thickens to an interval), on the arcs between consecutive zeros (Lemma 5.4), and on the cylinder ends (Theorem 6.1). The whole theorem can be organized around Lemma 5.3; what differs between the two regimes is the analytic driver that the separation feeds — the secular growth of Step 1 for time-independent mean profiles, the anchored cap of Lemma 5.2 in general.

Remark 5.9 (Conserved quantities; Cesàro-only convergence)*.* On the class (5), $\langle u^3 \rangle \equiv 0$ and the modal energy S is pointwise conserved (Lemma 3.1). Theorem 5.5 proves that the backward Cesàro mean of the flux exists and equals zero; no pointwise limit of $\Phi(\tau)$ as $\tau \to -\infty$ is asserted, and none holds in general. A rigorous oscillatory example (a $k = 1$ member of the polar sector (7)): take

$$A = 1 + \cos Z, \qquad \varphi_0 = 0, \qquad b = \delta\,(1 + \cos Z) \sin Z\, \sin \tau, \qquad 0 < |\delta| \le \tfrac{1}{2},$$

so that $B = \delta(1 + \cos Z) \sin Z\,(1 - \cos \tau)$. Both velocities are bounded uniformly in τ; every snapshot is a shear (a Z-dependent ξ-shift, isotopic to the identity) of the Proposition 7.1 portrait with parameter $\varepsilon = \delta \sin \tau$, $|\varepsilon| \le \frac{1}{2}$, hence satisfies Hypothesis 1.2; and by Lemma 3.4, $\Phi(\tau) = \frac{5\delta}{8} \sin \tau$: oscillatory, with zero Cesàro mean.

6. The cylinder

Theorem 6.1 (Cylinder version; time-independent mean profile)**.** *Let*

$$W = b(Z) + P_0(Z) \cos\bigl(k\xi + k\, b'(Z)\tau\bigr) + Q_0(Z) \sin\bigl(k\xi + k\, b'(Z)\tau\bigr)$$

on $\mathbb{T}_\xi \times \mathbb{R}_Z$, with b, P_0, Q_0 smooth and $S = P_0^2 + Q_0^2$ bounded (in the KNSS-symmetric setting of Corollary 1.4, boundedness of S is supplied by the u-bound via $\sup_\xi |u| = k\sqrt{S}$, and boundedness of b' by the v-bound via $\langle v \rangle_\xi = -b'$). If $C_v = \sup_{\tau \le 0} \|v(\cdot,\cdot,\tau)\|_{L^\infty} < \infty$ and, for a.e. $\tau \le 0$ — equivalently, by the shear conjugacy of Lemma 3.2, for one τ — no regular level component of $W(\cdot,\cdot,\tau)$ realizes the generator of $H_1(\mathbb{T}_\xi \times \mathbb{R}; \mathbb{Z}_2)$, then $b' \equiv 0$: the mean profile is constant. In particular the pointwise flux density vanishes, $\langle W_Z u^2 \rangle_\xi = \frac{k^2}{2}\, b' S \equiv 0$, and with it every average of it.

Proof. Steps 1–2 of Theorem 4.1 are local and apply verbatim: $b'' \equiv 0$ on $\Omega = \{S > 0\}$. Step 3 applies verbatim on any bounded gap; on an unbounded gap I the same argument works: the horizontal line $\{Z = Z^*\}$, $Z^* \in I$ with $b'(Z^*) \neq 0$, chosen at a Sard-regular value as before, is a regular level component (regular since $|\nabla W| = |b'(Z^*)| \neq 0$; a *complete* component since Z^* is an isolated root of $b(Z) = c^*$ by $b'(Z^*) \neq 0$, so the level contains no other nearby points) which is a circle in $\mathbb{T}_\xi \times \mathbb{R}$ generating $H_1(\mathbb{T}_\xi \times \mathbb{R}; \mathbb{Z}_2)$. The density argument of Step 4 then gives $b'' \equiv 0$ on $\mathbb{R}$, so $b' \equiv c_0$. It remains to exclude $c_0 \neq 0$ without periodicity: if $c_0 \neq 0$ then, with S bounded, $W = c_0 Z + O(1)$, so every regular value λ of sufficiently large absolute value has nonempty compact level set $\{W = \lambda\}$ contained in a slab and separating the two ends of $\mathbb{T}_\xi \times \mathbb{R}$ ($W - \lambda$ has opposite signs near the two ends, since $W = c_0 Z + O(1)$ with $c_0 \neq 0$). Lemma 5.3, applied on a slab containing this level, produces a component realizing the generator of $H_1(\mathbb{T}_\xi \times \mathbb{R}; \mathbb{Z}_2)$, violating the hypothesis. Hence $c_0 = 0$, i.e. $b' \equiv 0$; the flux statement follows from Lemma 3.4. ☐

Remark 6.2. We do not claim the time-dependent theorem on the cylinder. The per-arc machinery of Section 5 is local and applies to bounded components of $\{S > 0\}$, but the behavior at unbounded components requires additional end hypotheses (integrability of the weight and a far-field normalization of the averaged shear), and we leave the formulation and proof of an optimal cylinder statement for time-dependent mean profiles open.

7. Sharpness of the hypotheses

Proposition 7.1 (Dropping the v-bound)**.** *Let* $0 < |\varepsilon| \leq \frac{1}{2}$ *and let*

$$W_\varepsilon(\xi, Z, \tau) = b_\varepsilon(Z) + A(Z) \sin\bigl(\xi + b'_\varepsilon(Z)\,\tau\bigr), \qquad A = 1 + \cos Z, \quad b_\varepsilon = \varepsilon(1 + \cos Z) \sin Z$$

(the $k = 1$ solution (6) with $(P_0, Q_0) = (0,\, 1 + \cos Z)$, written in the polar form of Remark 3.3; here $S = A^2$). Then W, u, w are uniformly bounded on $\tau \leq 0$; all regular level components are contractible for all τ, so Hypothesis 1.2 holds — indeed in the stronger confinement form of Corollary 1.4; and

$$\bar{\Phi} = \frac{\varepsilon}{4\pi} \oint (\cos Z + \cos 2Z)(1 + \cos Z)^2 \, dZ = \frac{\varepsilon}{4\pi}\Bigl(2\pi + \frac{\pi}{2}\Bigr) = \frac{5\varepsilon}{8} \;\neq\; 0.$$

Here $A\, b''_\varepsilon \not\equiv 0$, so v grows linearly in $|\tau|$: the example satisfies every hypothesis of Theorem 4.1 except Hypothesis 1.1, and it is excluded exactly and only by that hypothesis.

Proof. Exactness of the solution is Lemma 3.2. For contractibility at $\tau = 0$, write the level structure explicitly: the level $\{W_\varepsilon(\cdot, \cdot, 0) = \lambda\}$ is $\{\sin \xi = g_\lambda(Z)\}$ with $g_\lambda := (\lambda - b_\varepsilon)/A$ on $\{A \neq 0\} = \mathbb{T} \setminus \{\pi\}$. Over each connected component of $\{|g_\lambda| < 1\}$ the level consists of the two ξ-branches $\arcsin g_\lambda$ and $\pi - \arcsin g_\lambda$, which join where $|g_\lambda| = 1$; a component of the level closes either as a *fold oval* (both ends of the Z-interval at the *same* sign of $g_\lambda = \pm 1$), which is contractible, or as a curve winding once in ξ (ends at *opposite* signs), which is not. Winding in Z is impossible for $\lambda \neq 0$: a component winding in Z surjects onto the Z-circle, hence meets $\{Z = \pi\}$, where W_ε vanishes identically, so any level meeting it has $\lambda = 0$. Now the envelopes factorize:

$$b_\varepsilon + A = (1 + \cos Z)\,(1 + \varepsilon \sin Z), \qquad b_\varepsilon - A = -(1 + \cos Z)\,(1 - \varepsilon \sin Z),$$

and for $|\varepsilon| \leq \frac{1}{2}$ both factors $1 \pm \varepsilon \sin Z$ are strictly positive, so $b_\varepsilon + A \geq 0 \geq b_\varepsilon - A$ with equality only at $Z = \pi$. Hence for $\lambda > 0$ every finite end of a component of $\{|g_\lambda| < 1\}$ has $g_\lambda = +1$ (the value $g_\lambda = -1$, i.e. $\lambda = b_\varepsilon - A \leq 0$, is impossible), and for $\lambda < 0$ symmetrically $g_\lambda = -1$: all ends carry the same sign, so every regular level component with $\lambda \neq 0$ is a fold oval, hence contractible (the two ξ-branches coalesce at $\xi = \pi/2$ where $g_\lambda = 1$, resp. $\xi = -\pi/2$

where $g_\lambda = -1$, and the defining Z-interval excludes π, so the closed curve is an embedded oval). It remains to dispose of $\lambda = 0$: since

$$W_\varepsilon(\xi, Z, 0) = (1 + \cos Z)\big(\sin\xi + \varepsilon \sin Z\big),$$

the zero level is the union of the seam $\{Z = \pi\}$ and the graphs $\{\sin\xi = -\varepsilon \sin Z\}$; on the seam $\nabla W_\varepsilon \equiv 0$ (both terms of each partial derivative carry the vanishing factor), and the closures of the graphs meet the seam (at $Z = \pi$ they reach $\sin\xi = 0$), so every connected component of the zero level contains critical points: $\lambda = 0$ is a critical value and its components are not regular level components — exempt under either reading of Remark 2.3. (For corroboration, the off-seam critical points of $W_\varepsilon(\cdot, \cdot, 0)$, at $\sin\xi = \pm 1$, reduce under $t = \tan(Z/2)$ to the cubics

$$t^3 + 3\varepsilon t^2 + t - \varepsilon = 0, \qquad t^3 - 3\varepsilon t^2 + t + \varepsilon = 0,$$

whose derivatives $3t^2 \pm 6\varepsilon t + 1$ have negative discriminant $36\varepsilon^2 - 12 \le -3$ for $|\varepsilon| \le \frac{1}{2}$: exactly one maximum and one minimum away from the critical circle, consistent with the pure-oval portrait; see Figure 1(a). For $|\varepsilon|$ large the positivity conclusion from the factorization fails, opposite-sign connections appear, and regular levels wind — the restriction on ε is what makes the example a hypothesis-respecting witness.) Contractibility is preserved for all τ because the evolution is the shear conjugacy H_τ of Lemma 3.2, a diffeomorphism of $\mathbb{T}^2$ isotopic to the identity. The flux integral is elementary: $b'_\varepsilon = \varepsilon(\cos Z + \cos 2Z)$ and $\oint(\cos Z + \cos 2Z)(1 + \cos Z)^2\, dZ = 2\pi + \frac{\pi}{2}$. The linear growth of v is the secular mechanism of (9): here $\sqrt{S}\,|b''_\varepsilon| = |A\, b''_\varepsilon| \not\equiv 0$, with predicted asymptotic slope $\max_Z |A\, b''_\varepsilon|$. The computational claims (PDE residual below 10^{-6} on a float64 grid, the exact flux value, the linear v-growth with fitted slope matching $\max_Z |A\, b''_\varepsilon|$) are checked by the ancillary script `counterexample_check.py` at $\varepsilon = 0.3$ (Appendix A); the contractibility claim is proved analytically above, not by the script. $\square$

Proposition 7.2 (Allowing horizontal winding: the compensated-gap witness)**.** *Let A be smooth, not identically zero, with $\operatorname{supp} A \subset S_0$ for a closed interval $S_0 \subsetneq \mathbb{T}$, and let V be a smooth periodic shear with $V \equiv c \ne 0$ on a neighborhood of S_0, $\oint_{\mathbb{T}} V\, dZ = 0$, and all the compensating variation of V located inside the gap $\mathbb{T} \setminus S_0$. Set $b(Z) := \int_0^Z V$ and let W be the $k = 1$ solution (6) with this time-independent b and $(P_0, Q_0) = (0, A)$, so $S = A^2$ (polar form of Remark 3.3, $\theta = \xi + b'(Z)\tau$). Then both velocity components of W are uniformly bounded (so Hypothesis 1.1 holds), and*

$$\bar{\Phi} = \frac{1}{4\pi} \oint b' A^2\, dZ = \frac{c}{4\pi} \oint A^2\, dZ \ne 0;$$

the hypothesis of Theorem 4.1 that fails is exactly Hypothesis 1.2: horizontal regular levels inside the gap realize the horizontal generator (Figure 1(c)).

Proof. b is smooth and periodic since $\oint V = 0$. On a neighborhood of $\operatorname{supp} A$ one has $b' = V \equiv c$, hence $b'' \equiv 0$ there, while $A \equiv A' \equiv 0$ outside S_0; so $A\, b'' \equiv 0$ on $\mathbb{T}$, the secular coefficient $\sqrt{S}\, b''$ vanishes identically, and $v = -b' - A' \sin\theta - A\cos\theta\, b''\tau = -b' - A'\sin\theta$ is bounded uniformly in τ, as is $u = A\cos\theta$: Hypothesis 1.1 holds. The flux value is Lemma 3.4 with $b'A^2 = cA^2$ supported in S_0. Since the flux is nonzero, Theorem 4.1 forces a violated hypothesis, and the only remaining candidate is Hypothesis 1.2; explicitly, V has zero mean and equals $c \ne 0$ near S_0, so $b' \ne 0$ somewhere in the interior of the gap $\{S = 0\}$, and exactly as in Step 3 of Theorem 4.1 a Sard-regular value there produces a horizontal regular level component $\{Z = Z^*\}$, realizing the horizontal generator of $H_1(\mathbb{T}^2; \mathbb{Z}_2)$. $\square$

Remark 7.3 (Hypothesis sharpness). *Drop Hypothesis 1.1*: Proposition 7.1 carries flux $5\varepsilon/8$ while satisfying Hypothesis 1.2 in its strongest (full-contractibility) form. *Drop Hypothesis 1.2*: Proposition 7.2 carries flux $\frac{c}{4\pi}\oint A^2\,dZ$ while satisfying Hypothesis 1.1. Thus each hypothesis of Theorem 1.3 is individually necessary — within every fixed harmonic-k stratum, by the covering transfer of Remark 7.4 below — and a fortiori neither hypothesis of Corollary 1.4 can be dropped.

Remark 7.4 (Sharpness at every harmonic). Both witnesses are stated at $k = 1$; the covering construction transfers them to every harmonic. For a harmonic-1 solution W_1 of (6) set

$$W_k(\xi, Z, \tau) := W_1(k\xi,\, Z,\, k\tau).$$

Then W_k is the harmonic-k solution (6) with the same (b, X_0): its phase is $k\xi + k\,b'(Z)\tau$. For the two witnesses of this section the status of each hypothesis is preserved (this is a statement about these witnesses, not about general members: regular components of $\cos(\xi - Z)$ have class $(1, 1)$, allowed by Hypothesis 1.2, while for its $k = 2$ cover $\cos(2\xi - Z)$ they have class $(1, 2) \equiv (1, 0) \bmod 2$, the forbidden horizontal generator). First, $v_k(\xi, Z, \tau) = v_1(k\xi, Z, k\tau)$, so $\sup|v|$ is unchanged (and $u_k = k\,u_1(k\xi, \cdot, k\tau)$ stays bounded). Second, at each time τ the level sets of $W_k(\cdot, \cdot, \tau)$ are the preimages of the level sets of $W_1(\cdot, \cdot, k\tau)$ under the k-fold cover $\eta_k(\xi, Z) = (k\xi, Z)$ of $\mathbb{T}^2$, with the same critical values (since $\nabla W_k = (k\,\partial_\xi W_1, \partial_Z W_1) \circ \eta_k$): a null-homotopic regular component lifts to k disjoint null-homotopic circles (a null-homotopic loop lifts to loops in any finite cover), while a horizontal circle $\{Z = Z^*\}$ is its own preimage. By Lemma 3.4 (same b and S, harmonic k) the flux gains exactly the factor k^2. Applied to Proposition 7.1 this produces, for every k, the witness $W_{\varepsilon,k} = b_\varepsilon + A\sin\bigl(k\xi + k\,b'_\varepsilon\tau\bigr)$ with $\bar\Phi = 5k^2\varepsilon/8$ and all regular level components still contractible; applied to Proposition 7.2, a witness with $\bar\Phi = \frac{k^2 c}{4\pi}\oint A^2\,dZ$, bounded velocities, and a horizontal regular level component. Hence the sharpness statements of Remark 7.3 hold within every fixed harmonic-k stratum, not merely for the union over k.

8. Open problems and scope

Open problem 8.1 (Multi-mode flux rigidity). Does the backward Cesàro limit (3) exist and equal zero for every ancient solution of (1) on $\mathbb{T}^2$ satisfying Hypotheses 1.1 and 1.2, without the single-mode restriction?

This is open to the best of the author's knowledge (note that even the existence of the limit is part of the question: in the multi-mode class $\bar\Phi$ may a priori be undefined), and the difficulty is structural rather than technical. On the single-mode classes the flux is a pure mean-shear channel (Lemma 3.4); in general the fluctuation channel $\langle W'_Z u^2\rangle$ is alive and the mean profile couples to an infinite hierarchy of wave interactions.

We also record the scope of the present result precisely: it concerns the single-mode strata only; the relation of the periodic-cell model to any primitive hydrodynamic system (e.g. axisymmetric Navier–Stokes in a thin-gap or long-wave regime) involves separate typing questions that are not addressed here, and no claim at the primitive level is made.

Appendix A. Machine verification

Four short standalone `Python` scripts accompany the paper as ancillary files of the arXiv submission (directory `anc/`, together with a `Makefile`, a short `README`, a pinned `requirements.txt`, and the checksum file `SHA256SUMS`; reference environment: Python 3.13.13, SymPy 1.14.0, NumPy 2.4.6). The scripts are citable as ancillary files of the arXiv submission;

a DOI deposit of the bundle is planned at publication. None is load-bearing for the proofs, which are complete above; they pin the computational identities and make the counterexample independently checkable in one command (`make verify`, run in the ancillary directory; `make verify-mutations` for the corruption battery). Each script asserts its checks and exits with nonzero status on any failure.

- `cas_class_exactness.py` — exact symbolic verification of Lemmas 3.1 and 3.4: the equivalence of the PDE residual with the rotation law (4) (both directions; symbolic in k), the exactness of the rotated solution (5), conservation of S, the velocity formula, the flux formula (8) and the parity identity $\langle u^3\rangle_\xi \equiv 0$ (at $k = 1, 2, 3$), and the compatibility of the Cartesian and polar forms (Remark 3.3): all exact CAS zeros.
- `counterexample_check.py` — float64 grid check of Proposition 7.1 at $\varepsilon = 0.3$ with no hand-coded derivatives (space derivatives by FFT collocation from sampled fields, time derivative by centered differences, predicted secular slope by SymPy differentiation). Reference run: calibration control ($b' \equiv 0$) passes its acceptance gate $|\bar\Phi| < 10^{-13}$ (the reference environment gives 7.8×10^{-21}; the exact value is platform-dependent float noise); sampled-field PDE residual 1.5×10^{-7}; time-mean flux $= 5\varepsilon/8$ to relative error 1.5×10^{-16}; fitted growth slope of $\sup|v|$ matching the predicted 1.390564 to relative error 2×10^{-5}.
- `verify_flux_formula.py` — independent symbolic checks of the flux formula in polar and Fourier form, the parity cancellation, and the exact counterexample integral $\oint(\cos Z + \cos 2Z)(1 + \cos Z)^2\, dZ = 5\pi/2$.
- `mutation_tests.py` — the reproducible corruption battery: six single-site mathematical corruptions of the checkers — including a drift of the parameter ε itself, caught by the gates anchored to literal reference constants — are applied to temporary copies, and each corrupted checker is asserted to exit nonzero.

Verification summary (reference run):

script	method	parameters	result (acceptance gate)
`cas_class_exactness`	symbolic	k symbolic / $k \le 3$	all residuals = 0 (exact zero)
`verify_flux_formula`	symbolic	—	all identities exact (exact equality)
`counterexample_check`	float64, FFT	256^2; slope 256×4096;	residual 1.5×10^{-7} ($< 10^{-6}$);
	collocation,	τ-step 2×10^{-4};	flux rel. 1.5×10^{-16} ($< 10^{-10}$, sprd $< 10^{-12}$);
	$\varepsilon = 0.3$ only	fit $[200, 400]$	slope rel. 2×10^{-5} ($< 10^{-2}$); cal. $< 10^{-13}$

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